SMART PATIO SHIELD

Technical Documentation

A Multimodal Machine Learning System for Nowcasting Wind-Driven Rain Events in Jamaica

The wet veranda that started it: wind-driven-rain blowing through the grillwork, exactly the event this system predicts.

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1. Introduction

a. Problem statement

This project develops a machine-learning system that predicts wind-driven rain events at specific outdoor locations in Jamaica, with sufficient reliability to trigger the automatic deployment of a protective screen over a covered, open-grille veranda. Wind driven rain (WDR), rain to which the wind imparts a horizontal velocity component, is an established subject of study in building science, where it is recognized as the most significant moisture source acting on building facades[1]. The problem is particularly acute in naturally ventilated tropical buildings, whose open corridors, balconies and terraces admit obliquely driven rain that a roof alone cannot exclude[2].

The veranda is roofed but enclosed by decorative metal grillwork rather than solid walls; consequently, ordinary vertical rainfall is excluded, but rain driven horizontally by gusting wind passes through the grille and wets the interior. The predictive target is therefore not rainfall in isolation but the *joint* occurrence of rain and wind sufficient to breach the veranda.

b. Aim and objectives

Aim. To design, implement, and evaluate a reproducible multimodal machine-learning pipeline that predicts wind-driven rain events from a combination of structured meteorological data and unstructured satellite imagery, and to quantify the predictive contribution of each data modality.

The specific objectives are:

- Acquire and document a reproducible, multi-year dataset of hourly weather observations and a corresponding archive of geostationary satellite imagery covering the study locations.
- Define an operationally meaningful binary target that captures the wind-driven-rain event, and engineer leakage-safe predictive features from the structured data.
- Train and rigorously evaluate three distinct model architectures – a tabular gradient-boosted baseline, a convolutional vision model and a multimodal fusion of the two – under an identical, honest evaluation protocol.
- Quantify the marginal contribution of each modelling choice and data modality through controlled ablation experiments.
- Produce a deployable inference artifact and the accompanying technical and management documentation.

[1] Blocken, B. & Carmeliet, J. (2004). A review of wind-driven rain research in building science. Journal of Wind Engineering and Industrial Areodynamics, 92(13), 1079-1130. https://doi.org/10.1016/j.jweia.2004.06.003

[2] Lim, C.H., Alkhair, M., Mirrahimi, S., Salleh, E. & Sopian, K. (2016). Optimization of wind driven rain barrier for tropical natural ventilated building. Research Journal of Applied Sciences, Engineering and Technology, 12(1), 69 – 85. The authors report that wind-driven rain wetted approximately 78% of the floor area of an open tropical corridor prior to mitigation.

c. Scope and framing

The system is framed honestly as a "*nowcaster*" rather than a long-range forecaster. It estimates whether the current hour constitutes a wet-veranda event – an hour in which rain, driven by wind, is likely to wet the interior of an open-grill veranda – given the current atmospheric state and its recent history. This framing is deliberate; it avoids overstating the system's capability and establishes a defensible foundation from which true forecasting extension (predicting the subsequent hour) follows naturally.

Beyond the academic deliverable, the system is designed with a commercial trajectory in mind: a localized rain-onset prediction service, exposed as an API, that could be licensed to outdoor-screen manufacturers, hospitality venues, agricultural operations, and other parties whose activities are exposed to sudden localized rainfall. This intent informs several design decisions (notably the per-location calibration and the reproducible, automatable data pipeline) but the commercial development itself is outside the scope of this document.

d. Course requirements addressed

The project is constructed to satisfy the requirements of COMP6830, which mandates a complete traversal of the data science process (retrieval from disparate sources, pre-processing and cleaning, preparation and modelling) using a combination of structured and unstructured data of non-trivial size. The mapping requirements to this work is summarized below.

Requirement	How it is satisfied
Disparate data sources	• Open-Meteo reanalysis API; • NOAA ONI climate index • NOAA GOES satellite archive on AWS
Structured + Unstructured	• Hourly tabular weather (structured) • multi-band satellite image patches (unstructured)
Relatively large	• 87,696 hourly tabular records • 87,648 multi-band image patches
Full data science process	Retrieval → cleaning → feature engineering → modelling → evaluation → deployment
3+ trained models	• XGBoost (tabular) • CNN (vision) • Multimodal fusion

2. System overview and methodology

a. Methodological framework

The project follows the CRISP-DM (Cross-Industry Standard for Data Mining) framework[3]. Business understanding is captured in the problem statement and target definition; data understanding in the exploratory analysis; data preparation in the feature engineering pipeline; modelling and evaluation in the three-model programme; and deployment in the inference artifact. The work proceeds iteratively, with findings at each stage informing revisions upstream.

b. Architecture at a glance

The system comprises a structured-data pipeline and an unstructured-data pipeline that converge at a fusion model, all sharing a single, consistent set of chronological data-split boundaries so that every model is evaluated on identical time periods.

> Structured pipeline: `fetch_weather.py` + `fetch_oni.py` → `build_features.py` → `baseline.py`
>
> Unstructured pipeline: `fetch_goes.py` → `build_labels.py` → `goes_dataset.py` → `cnn.py`
>
> Convergence: Model 3 consumes both the tabular and the satellite patches, joined on (location, timestamp).

c. The three models

Model	Modality	Architecture	Role
Model 1	Structured	XGBoost (+ logistic-regression sub-baseline)	Strong tabular baseline; establishes achievable performance
Model 2	Unstructured	ResNet-18 CNN (transfer learning)	Pure vision model on satellite patches
Model 3	Multimodal	Tabular branch + image branch, fused	Tests whether modalities are complementary

d. Reproducibility and engineering discipline

Reproducibility is treated as a first-class requirement rather than an afterthought. The pipeline is implemented in Python on the standard scientific stack[4], and the following practices are applied throughout:

- **Frozen data snapshots.** All acquisition scripts fix their date windows in code (1 May 2021 – 1 May 2026.

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[3] Wirth, R. & Hipp, J. (2000). CRISP-DM: Towards a standard process model for data mining. Proceedings of the 4th International Conference on the Practical Applications of Knowledge Discovery and Data Mining, 29-39.
[4] Core scientific stack: Harris, C.R. et al. (2020). Array programming with NumPy. Nature, 585, 357-362; Hoyer, S. & Hamman, J. (2017). Xarray: N-D labelled arrays and datasets in Python. Journal of Open Research Software, 5(1), 10; together with the pandas library for tabular manipulation.

- **Manifest receipts.** Every produced dataset is accompanied by a JSON manifest recording a SHA-256 hash of the file, the source, the parameters, and a timestamp.
- **Single source of truth for features**. All feature engineering is centralized in one module, so that training and future live inference apply identical transformations and no train/serve skew can arise.
- **Leakage-safe design.** Temporal features look only backward; the data split is strictly chronological; the test set is untouched until final evaluation.
- **Version control.** The full codebase is maintained under Git and hosted remotely with data and model binaries excluded by policy and only their manifests tracked.

3. Data Acquisition

Three disparate sources are combined: a reanalysis weather archive, a climate index, and a geostationary satellite archive. Each is retrieved by a dedicated, documented script that writes a reproducibility manifest.

a. Structured weather data (Open-Meteo/ECMWF IFS)

Hourly weather is retrieved from the Open-Meteo Historical Weather API[5], which serves ECMWF reanalysis data: EECWMF IFS at 9 km resolution for dates from 2017 onward, with ERA5 (25 km) as fallback for earlier dates[6]. The 1 May 2021 – 1 May 2026 window is therefore almost entirely by IFS. Reanalysis is a physics-based reconstruction of past weather: available observations (from stations, satellites, and other instruments) are assimilated into a numerical atmospheric model to produce a complete, internally consistent, gap-free gridded record. Its principal advantage for this project is completeness, with no missing hours across the five-year window, at the cost of being a model-derived product with known limitations (Section 4d).

Coverage. Two locations were chosen for their contrasting microclimates: Kingston (south coast) and Montego Bay (north coast). The window spans 1 May 2021 to 1 May 2026 at hourly resolution, yielding 1,827 days x 24 hours x 2 locations = 87,696 records, each with 20 meteorological variables.

Variable selection. Twenty hourly variables were requested, spanning temperature and humidity measures, precipitation, sea-level and surface pressure, cloud cover at four altitudes, wind speed/duration/gusts, evapotranspiration, vapour-pressure deficit, sunshine duration, and a categorical weather code. Each was selected for a documented meteorological reason relating to rainfall.

b. Climate context (NOAA ONI)

To provide season-scale context, the Oceanic Nino Index (ONI) is retrieved from NOAA. The ONI is the standard operational measure of the El Nino-Southern Oscillation (ENSO) – a slow, recurring cycle of sea-surface temperature in the tropical Pacific (typically 2-7 years per phase) that systematically biases Caribbean rainfall[7]. The published monthly index is retrieved, reshaped from its native wide year-by-season layout into a tidy monthly series, and each month is classified into an ENSO phase using the conventional ±0.5º threshold.

[5] Zippenfenig, P. (2023). Open-Meteo.com Weather API [Computer Software]. Zenodo. https://doi.org/10.5281/ZENODO.7970649

[6] Hersbach, H., Bell, B., Berrisford, P. et al. (2020). The ERA5 global reanalysis. Quarterly Journal of the Royal Meteorological Society 146(730), 1999-2049. https://doi.org/10.1002/qj.3803; ECMWF (2021). IFS Documentation CY47R3. European Centre for Medium-Range Weather Forecast, Reading, UK.

[7] Trenberth, K.E. (1997). The definition of El Nino. Bulletin of the American Meteorological Society, 78(12), 2771-2777. The ±0.5ºC threshold applied to the three-month running mean of Nino-3.4 sea-surface temperature anomalies is the operational NOAA definition.

c. Unstructured satellite imagery (NOAA GOES)

The unstructured modality is imagery from the GOES-East geostationary satellite, whose Advanced Baseline Imager (ABI) is a 16-band radiometer that images the full Western Hemisphere disk every ten minutes in its operational scan mode.[8] The complete historical archive is publicly hosted on AWS Open Data and requires no authentication.[9] This source was selected after determining that ground-level sky-camera collection could not yield a sufficiently large, labelled dataset within the project timeframe (Section 7a).

Three spectral bands are retrieve per hour and chosen to be complementary:

Band	Wavelength	Measures	Availability
C13 (clean IR)	10.3 µm	Cloud-top temperature	Day and night
C09 (water vapour)	6.9 µm	Mid-level atmospheric moisture	Day and night
C02 (red visible)	0.64 µm	High-resolution cloud structure	Daytime only

For each hour and location, a 64 x 64-pixel patch (approximately 128 x 128 km) is cropped around the city. Because the imagery is stored in a satellite's geostationary projection rather than geographic coordinates, each city's latitude and longitude are transformed into that projection using the parameters carried in each file's own metadata, so no pixel indices are hard-coded. The visible band, whose native resolution is finer (0.5 km versus 2 km for the infrared bands), is mean-downsampled to the common grid so that all three channels are spatially registered at 64 x 64. Patches are stored as compressed per-day archives keyed by location | ISO-timestamp, a key format that exactly matches the label table (Section 7b) so that images and labels join unambiguously. The transfer of GOES-East operations from the GOES-16 to the GOES-19 spacecraft in April 2025 is handled explicitly by selecting the correct archive bucket by date.

Acquisition engineering and completion. Full-disk files are large and downloading them in their entirety to extract a small patch would be prohibitive. The fetcher therefore opens each file lazily over the network and reads only the internal chunks intersecting the crop window, reducing transfer volume by orders of magnitude; it parallelizes the band downloads, is fully resumable, and proceeds in reverse-chronological order so that the most recent periods (the validation and test sets) are covered first. Acquisition is complete: all 1,826 days in the snapshot window were retrieved, totalling 87,648 patches. A dedicated integrity audit then scanned every patch for blank (failed) channels; affected days were deleted and re-fetched once, after which the identical gaps returned – confirming that the residual 0.29% of fully-blank patches correspond to genuine gaps in the public archive (hours with no scan available) rather than to recoverable download failures. All identified gaps fall in the training and validation periods; the held-out test period is complete.

[8] Schmit, T.J., Griffith, P., Gunshor, M.M., Daniels, J.M., Goodman, S.J. & Lebair, W.J. (2017). A closer look at the ABI on the GOES-R series. Bulletin of the American Meteorological Society, 98(4), 681-698. https://doi.org/10.1175/BAMS-D-15-00230.1

[9] NOAA GOES-R Series imagery, accessed via the Registry of Open Data on AWS (https://registry.opendata.aws/noaa-goes/). The archive is public and requires no authentication.

4. Exploratory Data Analysis

Before modelling, the structured data was examined systematically to understand its quality, structure, and the physical phenomena it encodes. The findings below drove a specific downstream design decision; they are reported in full in the accompanying EDA notebook.

a. Data quality

After cleaning (Section 4e), the dataset contains no missing values across all 87,696 records. Several variables are stored as integers (humidity, cloud cover, weather code, wind direction), producing a characteristic binning appearance in histograms; this was confirmed to be a display artifact rather than a data fault.

b. The daily rainfall cycle

Rainfall exhibits a strong daily rhythm, peaking sharply in the afternoon and early evening. This is the signature of tropical convection: daytime solar heating warms the land, which warms the air above it; the warm air rises, cools, and condenses into clouds that mature into afternoon showers. The pronounced afternoon maximum of convective rainfall over tropical land is a well-documented feature of the diurnal cycle[10]. The practical consequence is that the hour of the day is a genuinely predictive, motivating the temporal features in Section 5.

c. Spatial contrast and the rain shadow

Montego Bay is markedly wetter than Kingston – roughly 91% more mean rainfall. This is the rain-shadow effect: moisture-laden trade winds strike Jamaica's mountains and are forced upward on the windward north coast, where the air cools and releases its moisture near Montego Bay; by the time the air descends to leeward Kingston on the south coast, it has dried[11]. Location is therefore a genuine predictive feature, and the model learns two distinct rainfall regimes.

d. ENSO and the limits of reanalysis

Two findings here shaped the project materially. First, the expected ENSO relationship is present in the data: La Nina periods are wetter, consistent with the established Caribbean teleconnection[12], and the signal lives primarily in rainfall intensity rather than frequency – which is precisely the regime relevant to the wet-veranda event. However, at hourly

[10] Yang, G.-Y. & Slingo, J. (2001). The diurnal cycle in the tropics. Monthly Weather Review, 129(4), 784-801.
[11] Roe, G.H. (2005). Orographic precipitation. Annual Review of Earth and Planetary Sciences, 33, 645-671.
[12] Giannini, A., Kushnir, Y. & Cane, M.A. (2000). Interannual variability of Caribbean rainfall, ENSO, and the Atlantic Ocean. Journal of Climate, 13(2), 297-311; and Chen, A.A. & Taylor, M.A. (2002). Investigating the link between early season Caribbean rainfall and the El Nino + 1 year. International Journal of Climatology, 22(1), 87-106. Both report that the Caribbean late rainfall season tends to be drier in El Nino years and wetter in La Nina years.

granularity the ONI contributes little once weather-scale features are present, and it is retained as a low-cost seasonal conditioning variable rather than a strong predictor.

A documented limitation. The reanalysis data contains no thunderstorm weather-code across five years – climatological impossibility for Jamaica, which experiences frequent thunderstorms. This follows directly from how the product is constructed: at the reanalysis grid scale, deep convection is parameterized rather than explicitly resolved, so convective thunderstorms are not reliably diagnosed[13]. This is an inherent limitation of the structured modality, and it is one of the principal motivations for incorporating satellite imagery, which observes convective storm clouds directly.

e. Cleaning decisions

Three cleaning decisions were made, each documented with its rationale:

- boundary_layer_height was dropped owing to a six-month coverage gap that could not be reliably imputed.
- rain was dropped as a literal duplicate of precipitation (correlation 1.0).
- weather_code was excluded from the model features as a coarse, thunderstorm-blind derivative of information already present, though retained in the dataset for reference.

f. Target definition

The binary target wet_veranda encodes the wind-driven-rain event as a light-or-heavier rain combined with meaningful wind gusts, or heavier rain irrespective of wind:

```
wet_veranda = (precipitation > 0.1 mm AND wind_gusts > 15 km/h) OR (precipitation > 1.0 mm)
```

An honest disclosure follows from examining the target against the data. At the chosen thresholds, the target functions in practice as a rain detector rather than a distinct wind-driven-rain detector, i.e., it agrees with the simple condition that $precipitation > 0.1\ mm$ on 99.09% of hours. The wind-gust clause removes only about 11% of the rain-hours, and the heavy-rain clause contributes 159 of 8,053 positives. The clause is meteorological, in that, gusts during Jamaican rain-hours have a median of 28.4 km/h and a tenth percentile of 14.4 km/h, so a 15 km/h threshold sits below the bottom decile of the conditional distribution and filters almost nothing[14]. A sensitivity analysis (seen in the table below) shows the gust clause only begins to bind materially at much higher thresholds.

[13] At the 9 km grid scale of ECMWF IFS, deep convection is parameterized rather than explicitly resolved; convection-permitting simulation generally requires grid spacing of a few kilometres or less. Convective thunderstorm diagnosis is therefore not a reliable output of the product, which is consistent with the complete absence of thunderstorm weather-codes observed here.

[14] The wind-driven-rain threshold values are provisional and were not calibrated against a ground-truth observation log within the project timeframe; a threshold-sensitivity analysis characterizes their effect.

Gust threshold (km /h)	Positive rate	Agreement with precip > 0.1 mm	Positives from the light + gust clause
15 (used)	0.0918	0.9909	71.9%
25	0.0709	0.9700	63.7%
35	0.0481	0.9472	46.5%
45	0.0330	0.9321	22.0%

This disclosure matters because it interacts with satellite results (Section 8). The 0.1 mm precipitation thresholds places roughly three-quarters of the positive in the light-rain regime where infrared retrieval is physically weakest[15], which is central to why the vison modality and fusion behave as they do.

[15] Kidd, C. & Levizzani, V. (2011). Status of satellite precipitation retrievals. Hydrology and Earth System Sciences, 15(4), 1109-1116. Infrared methods infer rainfall from cloud-top temperature and systematically under-detect warm-topped (shallow) rainfall, which dominates tropical light rain.

5. Feature Engineering

All feature engineering is centralized in a single module (build_features.py) that serves as the single source of truth for both training and inference. The pipeline merges the climate index into the weather data, derives temporal and lag features, and constructs the target, writing one processed dataset (87,696 rows, 42 columns) plus a manifest.

a. Temporal features

From each timestamp, raw hour and month integers are derived (which tree models handle natively and which keep feature-importance interpretable), alongside cyclical sine/cosine encodings. The cyclical encoding preserves wrap-around adjacency that raw integer cannot express, i.e., hour 23 sits next to hour 0, and December next to January is lost in the raw form. Day-of-week is deliberately omitted; the EDA confirmed rainfall is flat across weekdays, as it must be, since the atmosphere has no concept of the working week.

b. Lag and trend features

Instantaneous snapshots miss the momentum and trajectory of the weather. Tel lag and trend features expose recent atmospheric history: prior-hour and three-hour precipitation and its rolling sum; prior-hour and three-hour maximum wind gust; three-hour and six-hour pressure trends; and lagged low-cloud and humidity terms. All are computed grouped by location, so one city's history never leaks into another's.

c. Leakage prevention

This is the most consequential aspect of the feature design. Data leakage, the inadvertent introduction of information into the training set that would not be available at prediction time, is a well-documented and frequently fatal failure mode in applied machine learning[16]. Because the target here is constructed from current-hour precipitation and wind gusts, those two variables would, if used directly as features, leak the answer's own ingredients into the prediction. Two safeguards prevent this. First, current-hour precipitation and gusts are excluded as features entirely; only their backward-shifted lag values are permitted, and the rolling windows are computed on shifted series so the current hour is never included in its own feature. Second, the feature pipeline is accompanied by a validation notebook that mechanically reconstructs each lag from raw data and asserts equality – for example, confirming that the three-hour precipitation sum at hour T equals exactly the sum of hours T-1 though T-3 and never includes T. Leakage is thus prevented by construction and verified by test, not merely assumed.

[16] Kaufman, S., Rosset, S., Perlich, C. & Stitelman, O. (2012). Leakage in data mining: Formulation, detection, and avoidance. ACM Transactions on Knowledge Discovery from Data, 6(4), 1-21.

d. The chronological split

The data is divided into training, validation, and test sets strictly by time. Random splitting of serially correlated time-series data leaks future information, because adjacent hours of the same weather system are distributed across both training and test sets, producing optimistically biased performance estimates; blocked, chronological evaluation is the appropriate protocol[17]. The split assigns the earliest 70% of the window to training, the next 15% to validation, and the most recent 15% to test, applying the same calendar boundaries to both locations.

Split	Records	Period (approx.)	Positive rate
Train	61,386	May 2021 – Oct 2024	9.02%
Validation	13,154	Oct 2024 – Jul 2025	9.10%
Test	13,156	Jul 2025 – May 2026	10.03%

The near-constant positive rate across splits confirms no split landed on an anomalously wet or dry stretch. Hurricane Melissa – the Category 5 storm that made landfall in western Jamaica on 28 October 2025, the strongest on record to strike the island[18] – falls within the test period; final results will therefore be reported both including and excluding the Melissa window, so that a single extreme event neither flatters nor distorts the reported performance. These same split boundaries are inherited unchanged by the image and fusion models, so that all three models are compared on identical time periods.

[17] Bergmeir, C. & Benitez, J.M. (2012). On the use of cross-validation for time series predictor evaluation. Information Sciences, 191, 192-213.

[18] National Hurricane Center (2026). Tropical Cyclone Report: Hurricane Melissa (AL132025). NOAA/National Weather Service. Melissa made landfall in western Jamaica on 28 October 2025 at an estimated 160 kt (Category 5) with a minimum central pressure of 892 hPa – the strongest hurricane on record to make landfall in Jamaica.

6. Model 1 – Tabular Baseline

a. Evaluation metric

The headline metric throughout is PR-AUC (the area under the precision-recall curve, equivalently average precision). Given the ~9% positive rate, accuracy is misleading – a model that always predicts "dry" achieves 91% accuracy while being completely useless. Under strong class imbalance, ROC curves can also present an over-optimistic picture because the large negative class renders the false-positive rate insensitive to substantial numbers of false alarms; the precision-recall plot is the more informative choice in this regime[19]. PR-AUC summarizes the trade-off between *precision* (of the hours flagged wet, how many truly were) and *recall* (of the truly wet hours, how many were caught) across all decision thresholds. A random classifier scores approximately the positive rate (~0.09), so the metric is self-calibrating.

b. Models and training

Two models were trained on identical data. A logistic regression with standardized features and balanced class weights[20] serves as a linear reference floor. An XGBoost gradient-boosted tree ensemble[21] serves as the strong baseline; it applies class imbalance weighting (scale_pos_weight) and, critically, early stopping – training halts when validation PR-AUC ceases to improve[22]. Without early stopping, training PR-AUC was observed to climb toward 0.95 while validation PR-AUC peaked near 0.72 and then decayed, the classic signature of overfitting; early stopping selects the genuinely best model (around boosting iteration 174). Class imbalance is addressed by cost weighting rather than by resampling[23].

c. Results

On the validation set, XGBoost outperforms logistic regression at every operating point, and both vastly exceed the random floor:

Model	Features	Validation PR-AUC
Random baseline	-	0.091
Logistic regression	All (linear)	0.628
XGBoost (early-stopped)	All (non-linear)	0.723

[19] Saito, T. & Rehmsmeier, M. (2015). The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. PLOS ONE, 10(3), e0118432; and Davis. J. & Goadrich, M. (2006). The relationship between precision-recall and ROC curves. Proceedings of the 23rd International Conference on Machine Learning, 233-240.
[20] Pedregosa, F. et al. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825-2830.
[21] Chen, T. & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 785-794.
[22] Prechelt, L. (1998). Early stopping – but when? In Neural network Tricks of the Trade (LNCS 1524), 55-69. Springer.
[23] Synthetic oversampling (e.g. Chawla, N.V., Bowyer, K.W., Hall, L.O. & Kegelmeyer, W.P. (2002). SMOTE: Synthetic minority over-sampling technique. Journal of Artificial Intelligence Research, 16, 321-357) was deliberately not used: interpolating synthetic observations into a time series fabricates atmospheric states that never occurred and disrupts temporal structure.

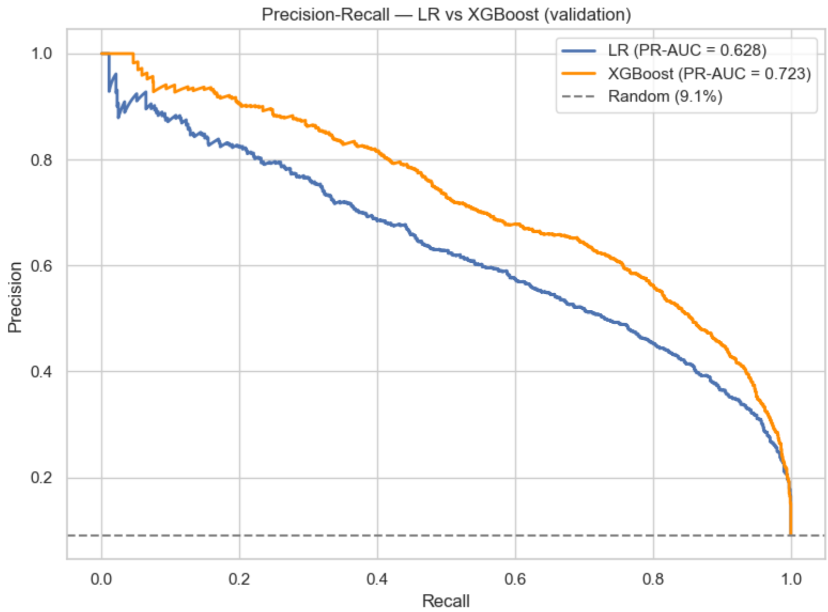

Figure 1. Validation precision-recall curves. XGBoost (orange) dominates logistic regression (blue) across the entire recall range; both far exceed the random baseline.

d. What the model relies on

Feature-importance analysis shows a single feature – prior-hour precipitation (precip_lag1) – accounting for roughly 42.5% of the model's decision weight, some four times the next feature. This reflects the strong hour to hour persistence of weather. Cloud cover (at mid and low altitudes) provides the strongest anticipatory signal; the daily-cycle features capture convection timing; and the location indicator confirms the rain-shadow effect is real model signal. Pressure-trend and ENSO features rank low – a finding that was investigated in detail below.

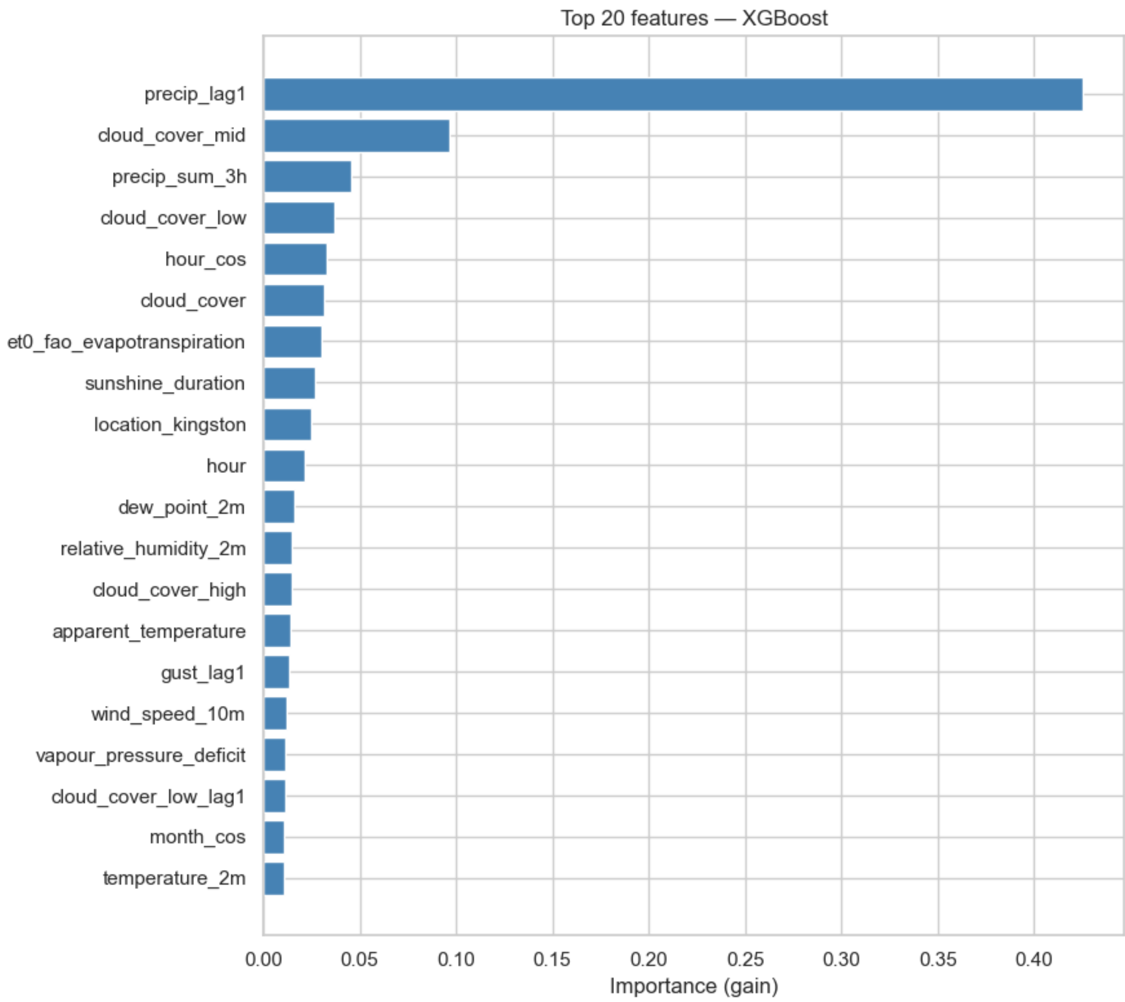

Figure 2. XGBoost feature importance (top 20, by gain). Prior-hour precipitation dominates, followed by cloud cover and the daily-cycle terms.

e. Persistence decomposition

Persistence, the assumption that current conditions continue, is a standard reference forecast in meteorology against which any proposed model must demonstrate genuine skill[24]. To answer the natural sceptical question, *"Is the model merely exploiting persistence?"*, a reference model was trained using only prior-hour precipitation. It achieves PR-AUC 0.459. Measured against the random floor, persistence therefore accounts for approximately 58% of the achievable lift, and the engineered features supply the remaining 42%. The precision-recall curves show where that engineered contribution lies: at higher recall. Persistence sustains precision only at low recall; it can confirm rain that is already falling, but it is structurally blind to rain that is just beginning. The engineered features are what enable the model to anticipate rain onset rather than merely detect its continuation. The model is therefore anchored on persistence but is substantially and measurably more than a persistence model.

[24] Persistence is a standard reference forecast in meteorology, against which a proposed model must demonstrate genuine skill; see Murphy, A.H. (1992). Climatology, persistence, and their linear combination as standards of reference in skill scores. Weather and Forecasting, 7(4), 692-698.

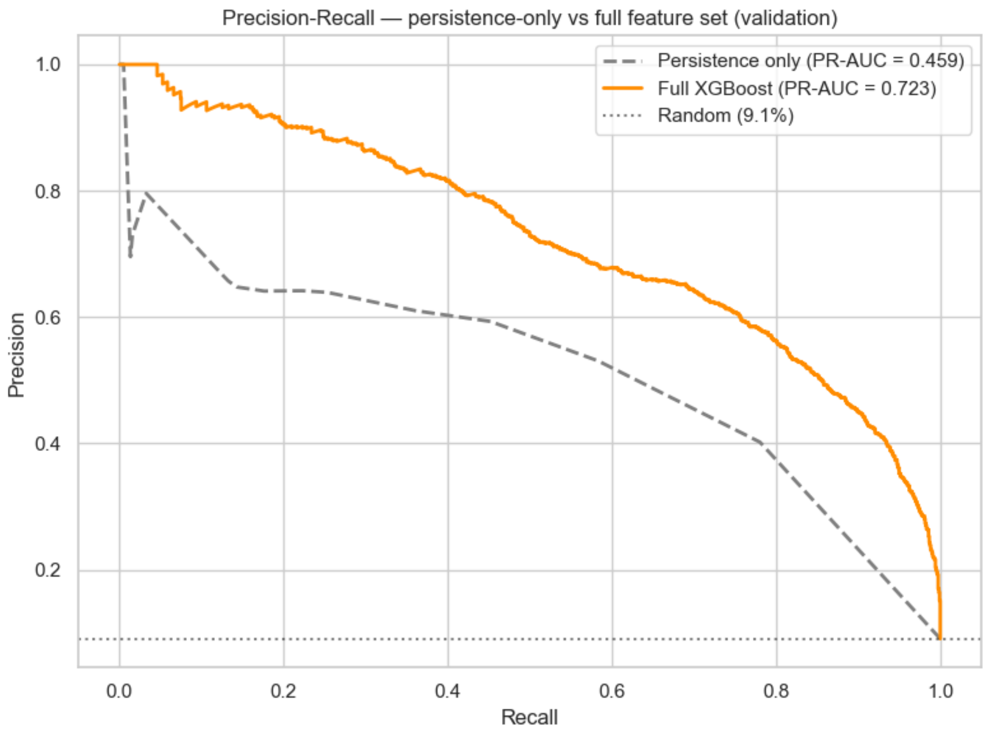

Figure 3. Persistence-only model (grey, PR-AUC 0.459) versus the full feature set (orange, 0.723). The widening gap at higher recall is the rain-onset regime, where the engineered features earn their value.

f. Pressure-trend investigation

Because pressure tendency is a classical storm precursor yet ranked low in importance, it was investigated directly. Three findings emerged. First, the storm signal is genuinely present: wet hours show a median pressure fall of roughly 0.8-1.0 hPa over the preceding 3-6 hours, versus a slight rise in dry hours, and this holds in the median (so it is not an artifact of Hurricane Melissa). Second, the signal is redundant: removing the pressure-trend features changes validation PR-AUC by only + 0.004. because the precipitation-lag features already carry the same information more directly. Third, the raw trend is confounded by the atmospheric semidiurnal tide – a solar-driven twice daily oscillation in surface pressure whose amplitude is largest in the tropics[25], and which appears in the Jamaican data as a regular ~2.8 hPa daily swing peaking near 10am and 10pm: large enough to mask the ~1 hPa weather signal. The features were retained as low-cost inputs; the investigation illustrated the distinction between a feature being scientifically real and being marginally useful to a model that already possesses a stronger correlate.

[25] Dai, A. & Wang, J. (1999). Diurnal and semidiurnal tides in global surface pressure fields. Journal of Atmospheric Sciences, 56(22), 3874-3891; the classical treatment is Chapman, S. & Lindzen, R.S. (1970). Atmospheric Tides: Thermal and Gravitational. D. Reidel. Tidal pressure amplitudes are largest in tropics, which is why the effect is pronounced in the Jamaican data.

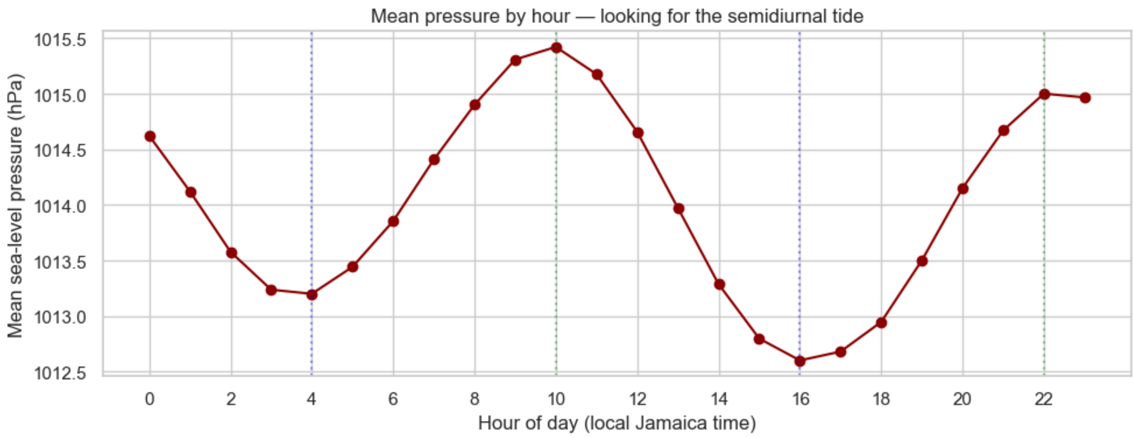

Figure 4. The semidiurnal atmospheric pressure tide in the Jamaican data: a regular twice-daily oscillation that confounds raw pressure-trend features.

g. Summary of Model 1

Model 1 is complete and validated. The tabular XGBoost baseline achieves validation PR-AUC 0.723, strictly dominating the logistic-regression floor (0.628) and the random baseline (0.091). Every claim about the model is supported by a controlled experiment: early stopping prevents overfitting; the persistence decomposition quantifies the 58/42 split between weather inertia and engineered features; and the pressure-trend investigation demonstrates disciplined feature evaluation. The held-out test set remains untouched, reserved for final evaluation in Section 9.

7. Model 2 – Vision Model

a. Rationale for the satellite modality

The structured data is, by construction, a summary of the atmosphere – single numeric values per hour. The satellite provides a direct observation of the cloud field. The distinction is decisive in two ways. First, it fills a documented blind spot: the reanalysis cannot represent thunderstorms (Section 4d), whereas infrared imagery observes connective storm clouds directly. The relationship between cold infrared cloud tops and convective rainfall is long-established and underpins satellite precipitation estimation[26]. Second, scalar summaries discard spatial structure that images retain – two hours with identical cloud-coverage percentages may correspond to a harmless overcast or to an organized storm cell approaching the city, a difference invisible to the tabular row but plain in the image.

A feasibility study confirmed that the signal is present in the imagery, Infrared Band 13 cleanly distinguishes a clear day (warm cloud tops), an ordinary rain hour (cooler, textured tops), and Hurricane Melissa (cloud tops at 191 K, around -82ºC). An important caveat was identified at the same time, and it is a known structural limitation of the technique rather than an artifact of this dataset: because infrared methods infer rainfall from cloud-top temperature, they systematically under-detect warm-topped (shallow) rain, which is common in the tropics[27]. Ordinary Jamaican rain is therefore only subtly cooler than clear sky in the infrared – which is itself the argument for a multi-band, multimodal approach rather than infrared vision alone, and which anticipates the operating character measured in Section 7d.

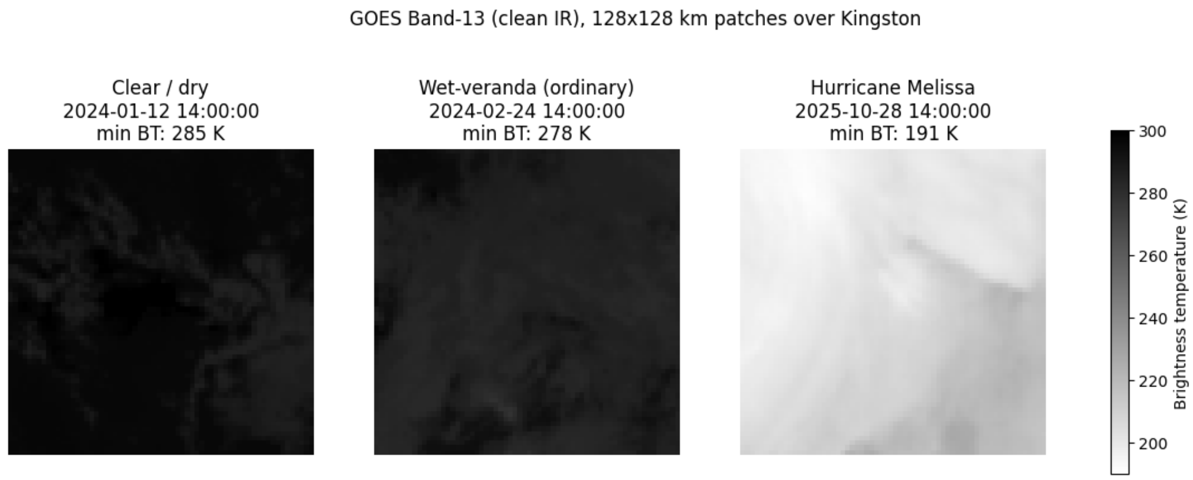

Figure 5. Feasibility study: GOES Band-13 infrared over Kingston for a clear day, an ordinary rain hour, and Hurricane Melissa. The convective signal is clearly present, most dramatically for the Category05 storm.

[26] Arkin, P.A. & Meisner, B.N. (1987). The relationship between large-scale convective rainfall and cold cloud over the Western Hemisphere during 1982-1984. Monthly Weather Review, 115(1), 51-74. This established the basis for inferring convective rainfall from cold infrared cloud tops.
[27] Kidd, C. & Levizzani, V. (2011). Status of satellite precipitation retrievals. Hydrology and Earth System Sciences, 15(4), 1109-1116. Infrared methods infer rainfall from cloud-top temperature and therefore systematically under-detect warm-topped (shallow) rainfall while over-detecting cold, non-precipitating hours.

b. Bridging images to labels

A label table (build_labels.py) maps every (location, timestamp) to its wet_veranda label and its split membership, using the identical chronological boundaries as Model 1. Each row's key is formatted to match the keys inside the satellite archives exactly, so that the image-loading dataset can join each patch to its label and split unambiguously. This bridge is what guarantees that the vision and fusion models are evaluated on precisely the same periods and labels as the tabular model. The label table reproduces the Model 1 split exactly: 61,386 / 13,154 / 13,156 records at 9.02% / 9.10% / 10.03% positive. With acquisition complete, the image-label join covers essentially the full dataset: 61,386 training and 13,154 validation patches were available for model development, and 13,108 test patches for final evaluation.

c. Architecture and training

Model 2 is a ResNet-18 convolutional network[28], pretrained on ImageNet[29] and adapted by transfer learning. Training a network from scratch on the available patches would overfit; instead, the pretrained network (whose early convolutional layers encode general-purpose visual primitives such as edges and textures that transfer across domains[30]) is fine-tuned end-to-end on the satellite data. Two adaptations are made. First, the initial convolutional layer is replaced to accept the satellite channel count (two for infrared-only, three with the visible band) rather than the three RGB channels; its weights are initialized by averaging the pretrained filters across the colour dimension and replicating the result across the input channels, so that the learned low-level detectors are preserved rather than discarded, and every spectral band is examined from the first training step by the full battery of pretrained filters. Second, the final fully-connected layer is replaced with a single-logit binary classification head. ResNet-18 (rather than a deeper variant) was chosen to match model capacity to the modest patch size (64 x 64) and dataset scale.

Training deliberately mirrors the tabular discipline: a class-weighted loss up-weights the minority wet class (positive-class weight ≈ 10, the neural-network analogue of the tree model's imbalance weighting); optimization uses Adam[31] at a deliberately low learning rate (1×10^{-4}), appropriate for fine-tuning pretrained weights rather than training from random initialization; training is early-stopped on validation PR-AUC, retraining the best-scoring weights; and PR-AUC is the headline metric, so that Model 2's result sits in the identical frame as Model 1's. The implementation uses PyTorch[32] and was trained on a cloud GPU

[28] He, K., Zhang, X., Ren, S. & Sun, J. (2016). Deep residual learning for image recognition. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 770-778.
[29] Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K. & Fei-Fei, L. (2009). ImageNet: A large-scale hierarchical image database. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 248-255.
[30] Yosinski, J., Clune, J., Bengio, Y. & Lipson, H. (2014). How transferable are features in deep neural networks? Advances in Neural Information Processing Systems, 27, 3320-3328. Early convolutional layers learn general-purpose features (edges, textures) that transfer across domains, while later layers grow increasingly task-specific.
[31] Kingma, D.P. & Ba, J. (2015). Adam: A method for stochastic optimization. International Conference on Learning Representations.
[32] Paszke, A. et al. (2019). PyTorch: An imperative style, high performance deep learning library. Advances in Neural Information Processing Systems, 32, 8024-8035.

(Tesla T4), with per-channel normalization statistics computed on the training split only and reused unchanged for validation and test. Because GPU neural-network training is not bit-for-bit deterministic, exact reproducibility is not guaranteed; the model manifest therefore records the outcome (metrics and full training history) as the reproducibility receipt, and run-to-run stability was verified empirically (Section 7d).

d. Training behaviour and results

Training on the full 61,386-patch training split exhibited a consistent and informative pattern: validation PR-AUC peaked within the first one to three epochs and declined thereafter, while training loss continued to fall steeply; this is the signature of a model that extracts the available generalizable signal almost immediately and then memorizes training-specific detail. Early stopping captured the peak in every run. Two checks were performed to establish that this early peak reflects the problem rather than a tuning artifact. A reduced learning rate ($3\ x\ 10^{-5}$ in place of $1\ x\ 10^{-4}$) produced a lower peak (0.360), ruling out learning-rate overshoot as the cause. And a full repeat of the selected configuration under a fresh random initialization reproduced the result to within ±0.002 (0.450 versus 0.448), establishing that the reported numbers are stable rather than fortunate draws of GPU non-determinism.

Model selection followed the standard protocol. The configuration and run with the best validation PR-AUC (infrared + visible, 0.450) was selected, and only that model was evaluated once on the held-out test set. It achieved test PR-AUC of 0.440, in close agreement with its validation score, indicating honest generalization with no meaningful validation overfitting, At the default 0.5 decision threshold, the model catches 1,114 of 1,315 true wet-veranda hours (recall 0.85) while raising 2,882 false alarms (precision 0.28). This operating character (high recall, low precision) is precisely the profile anticipated in Section 7a from the physics of infrared retrieval; the model fires reliably on cold-cloud convective signatures but cannot finely separate borderline warm-rain hours from dry ones, and so over-triggers. The decision threshold is tuneable along the precision-recall curve; the appropriate operating point is a deployment decision governed by the relative cost of a missed event versus a false alarm.

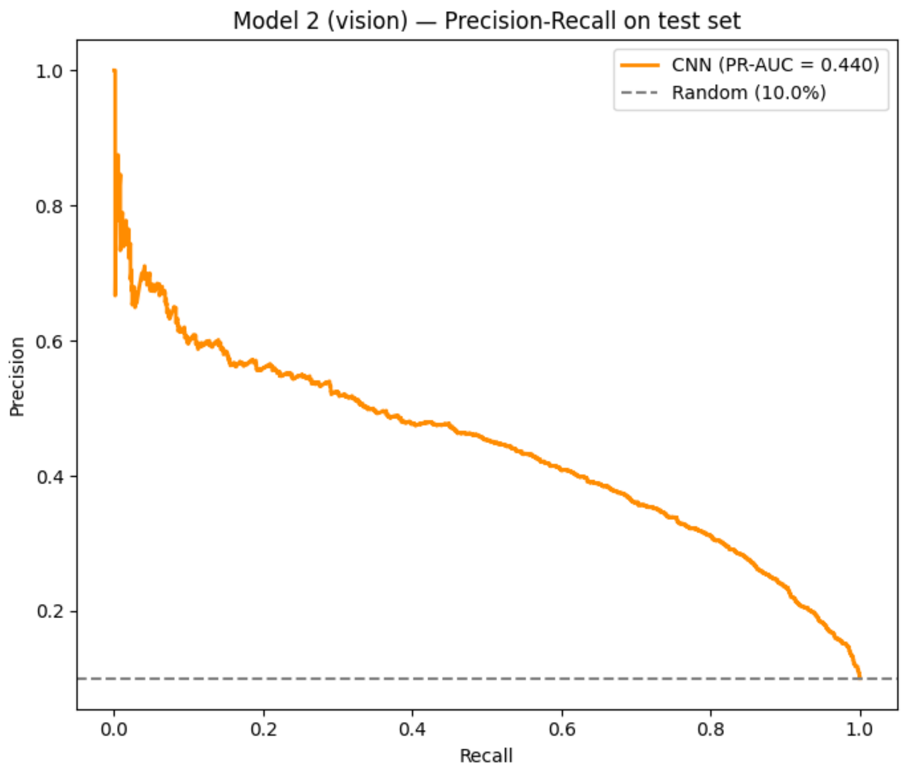

Figure 6. Model 2 (vision, IR + visible) precision-recall on the held-out test set: PR-AUC 0.440 against a 10.0% random floor. The curve ramains well above random across the full recall range - a real but modest signal, consistent with the physical limits of infrared imagery for warm-rain detection.

i. The right baseline for the vision model: climatology

Model 2's test PR-AUC of 0.440 must be read against the correct reference. The random floor (the positive rate, ~0.10) flatters the model, because the label has a strong daily cycle, i.e., the positive rate swings roughly seventeen-fold from about 1.8% at 06:00 to 31.1% at 14:00. The appropriate floor is therefore climatology: the base rate conditioned on hour, month, and location fit on the training split alone[33]. Evaluated on the same held-out test hours as the CNN:

Reference / model	Test PR-AUC
Random floor (positive rate)	0.1003
P(wet | hour)	0.2669
P(wet | hour, month, location)	0.2933
P(wet | hour, month, location) climatology floor	0.3195
Model 2 (ResNet-18 on GOES)	0.4397
Model 1 (XGBoost tabular)	0.7557

[33] Climatology here is the empirical base rate conditioned on hour-of-day, month, and location, P(wet | hour, month, location), estimated on the training split only and applied unchanged to validation and test. It is the natural reference for a model whose label has a strong diurnal and seasonal cycle.

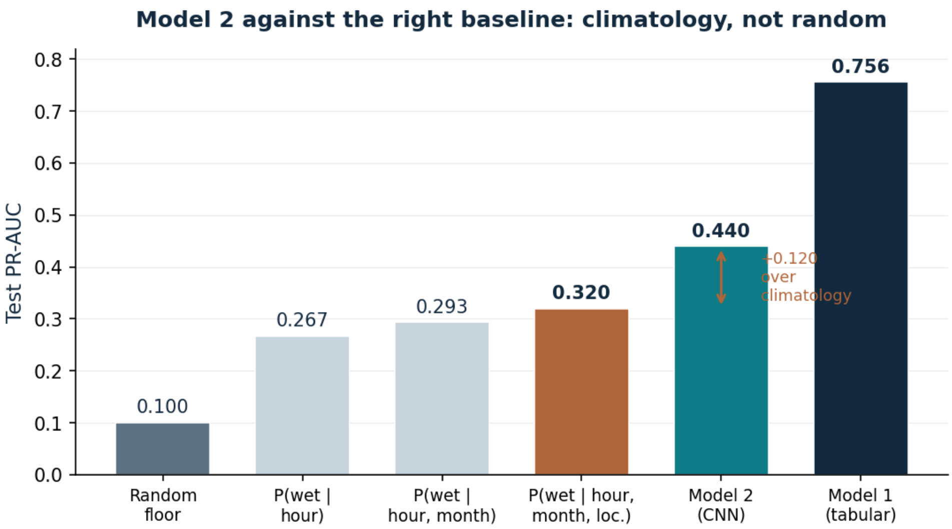

Figure 7. Model 2 against the correct baseline. Five years of GOES imagery through a ResNet-18 add +0.12 PR-AUC over a three-column climatology lookup table - an honest measure of the vision model's contribution, far smaller than its +0.34 margin over the random floor would suggest.

Read honestly, the CNN's genuine margin over is +0.12 over climatology, not +0.34 over random. The vision model does not extract real information beyond the diurnal and seasonal cycle, but the margin is modest; a framing that is both more defensible and more informative than the raw number, and one that directly motivates the fusion analysis in Section 8.

e. Channel ablation: the contribution of the visible band

To quantify what the visible band adds, two models were trained under otherwise identical settings: one on the two infrared channels alone, and the other on all three bands. This is the vision-model analogue of the persistence decomposition: a single controlled comparison isolating one design choice, served by one acquisition (all bands are fetched once), so the ablation is a training-time switch rather than a second download. The infrared-only configuration reached validation PR-AUC 0.427; adding the visible band raised it to 0.450, a contribution of +0.023. The gain is real but bounded by a structural constraint identified in advance, i.e., the visible band carries information only in daylight (at night it records near-zero reflectance), so it can assist only the roughly half of events that occur during the day. Consistent with the band supplying genuine structure rather than noise, the three-channel model also trained more gradually, peaking at epoch 3 rather than epoch 1. The three-channel configuration was selected as the definitive Model 2 and as the vision branch for fusion.

Configuration	Channels	Validation PR-AUC	Test PR-AUC
Infrared only	C13 + C09	0.427	- (not evaluated; selection by validation)
Infrared + visible (selected)	C13 + C09 + C02	0.450	0.440

i. Channel ablation: a day/night caveat on the visible band

Section 7e attributed a +0.023 validation gain to adding the visible band (C02). A confound must be disclosed. The 0.64 µm visible reflectance is near-zero at night and rises through the day, so it is in part a solar-zenith-angle (time-of-day) signal rather than pure cloud reflectance[34]. Because the label's base rate varies seventeen-fold across the day, a channel that encodes time-of-day is itself weakly predictive. The ablation as run cannot separate the idea that the visible band sees bright convective clouds from the visible band tells the model it is mid-afternoon. The +0.023 is therefore reported with the caveat that some portion may be a temporal proxy; a clean separation (a daytime-only evaluation, or supplying the infrared-only model with explicit time-of-day planes) is identified as future work.

f. Summary of Model 2

Model 2 is complete and validated. The satellite vision model achieves a test PR-AUC of 0.440 – roughly 4-5 times the random floor of 0.10, and reproducible across independent training runs (±0.002). The channel ablation attributes +0.023 of the validation performance to the visible band. Vision alone underperforms the tabular baseline (0.723 validation), and its errors have a distinctive character: high recall on convective events, low precision on ambiguous warm-rain hours. This is the empirical foundation of the fusion hypothesis, i.e., the two modalities fail on different hours, and Section 8 tests whether combining them covers each other's blind spots.

[34] The 0.64µm visible reflectance depends on solar illumination and is therefore near-zero at night; it functions in part as a solar-zenith-angle (time-of-day) indicator. Because the label's base rate varies roughly seventeen-fold across the day, a time-of-day signal is itself weakly predictive, so the visible-band's measured contribution cannot be attributed purely to cloud reflectance without a daytime-only control.

8. Model 3 – Multimodal Fusion

Model 3 combines the tabular and satellite modalities. The premise was that their errors are complementary, so combining them should improve on either alone. This section reports the outcome across five fusion protocols and a mechanistic explanation grounded in the physics of infrared rainfall retrieval. The headline result is a rigorously established negative one: no fusion variant beats the tabular baseline at this scale, and the reason is specific and defensible.

a. Fusion designs and protocols tested

Fusion was evaluated five ways, progressing from simplest to most careful, so that a null result could not be attributed to an inadequate protocol. Evaluation protocols:

- **Late fusion (raw).** A logistic meta-learner over the two models' output probabilities, fit on the validation set.
- **Late fusion (calibrated).** As above, with each base model's probabilities isotonically calibrated first[35].
- **Feature-level fusion.** The CNN's 512-dimensional embedding concatenated with the tabular features, with a joint head.
- **Out-of-fold stacking.** The meta-learner refit on out-of-fold tabular predictions across the training period (~51,000 rows) rather than on the smaller, optimistically biased validation set[36].
- **Intensity-conditional fusion.** The meta-learner given interaction terms between the vision probability and convective-context signals (humidity, low cloud), so it could weight the satellite differently in likely heavy-rain conditions.

b. Results: no variant beats the tabular baseline

On the held-out test set (13,108-hour intersection where both modalities are present), the ordering is unambiguous:

Model/ fusion variant	Test PR-AUC	Recall	Precision
Model 1 – Tabular (baseline)	0.7557	0.841	0.538
Late fusion – calibrated (best fusion)	0.7406	0.856	0.527
Late fusion – raw	0.7342	0.891	0.482
Feature-level fusion	0.6660	0.827	0.448
Intensity-condition fusion (OOF)	0.6622	0.840	0.413
Out-of-fold late fusion	0.6523	0.842	0.402
Model 2 – vision alone	0.4397	0.847	0.279

[35] Isotonic regression is monotone and therefore cannot change a single model's ranking-based PR-AUC; the small drop observed when calibrating Model 1 in isolation is an artifact of ties introduced by the fitted step function, and is not a genuine performance change.
[36] Stacking (stacked generalization): Wolpert, D.H. (1992). Stacked generalization. Neural Networks, 5(2), 241-259. A meta-learner is trained on base-model predictions; to avoid optimistic bias it should be trained on out-of-fold predictions rather than in-sample ones.

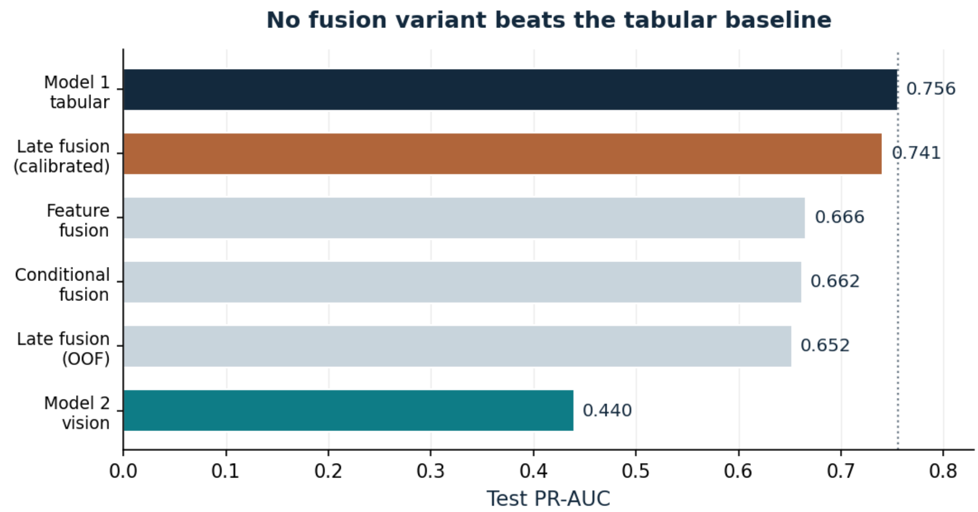

Figure 8a. Final model comparison on the held-out test set. Every fusion variant falls below the tabular baseline (dotted line). The best fusion result (calibrated late fusion at 0.741) still trails Model 1's 0.756.

Two protocols observations are worth recording. First, out-of-fold stacking made fusion worse, not better (0.652). The meta-learner's fitted weights, [tabular 3.41, vision 6.38], reveal why. On the training folds, the CNN's in-sample predictions looked artificially strong, so the meta-learner over-trusted the vision branch, which then hurt on test data. This is the expected consequence of stacking with out-of-fold predictions on one branch but in-sample predictions on the other. A fully clean version would require k-fold retraining of the CNN, which was outside the timeframe. Second, intensity-conditional fusion moved the needle only slightly (0.662). The context-interaction weights were correctly signed (positive on humidity and low cloud, i.e. it should trust vision more when the atmosphere looks convective) but far too small to overcome the weakness of a 0.44 vision branch. Both variants are reported as robustness checks that confirm the negative finding rather than as improvements.

A follow-up diagnostic isolated why feature-level fusion in particular underperformed. Scale mismatch between the modalities was not the cause: the concatenated matric is standardised before the joint head is fitted. Two other factors were. First, the joint is linear, whereas the tabular model's strength is non-linear. Replacing the logistic head with a gradient-boosted one raises feature fusion from 0.666 to 0.715. Second, the 512-dimensional image embedding outnumbers the roughly 35 tabular columns under a uniform penalty, and the tabular signal is diluted rather than complemented; tabular features alone under the same linear head score 0.680, above the 0.666 obtained when the embedding is added. Compressing the embedding to sixteen principal components, which retain 97.8% of its variance, recovers this to 0.688 and incidentally shows the vision representation to be far lower-rain than its nominal dimensionality. Refitting the same head on the larger out-of-fold set made matters worse (0.625), consistent with the in-sample character of the CNN's training-hour embeddings already noted above. None of these variants reachers the tabular baseline of 0.756, so the conclusion is unchanged. The original figure understated feature-level fusion by conflating it with the choice of head.

c. Why fusion fails: conditional informativeness, not redundancy

The natural explanation that the satellite is simply redundant with ERA5's cloud fields is wrong, and the correct explanation is more interesting. Measuring the discrimination of a single satellite summary (minimum cloud-top brightness temperature) against a single ERA5 column (low-cloud cover), stratified by rain intensity, shows that the satellite is conditional informative.

Rain intensity (mm/h)	Satellite AUC (-min BT)	ERA5 low-cloud AUC	Winner
0.1 – 0.5	0.635	0.789	ERA5
0.5 – 1	0.670	0.797	ERA5
1 – 2	0.762	0.798	ERA5 (≈)
2 – 5	0.839	0.786	Satellite
>5	0.919	0.817	Satellite

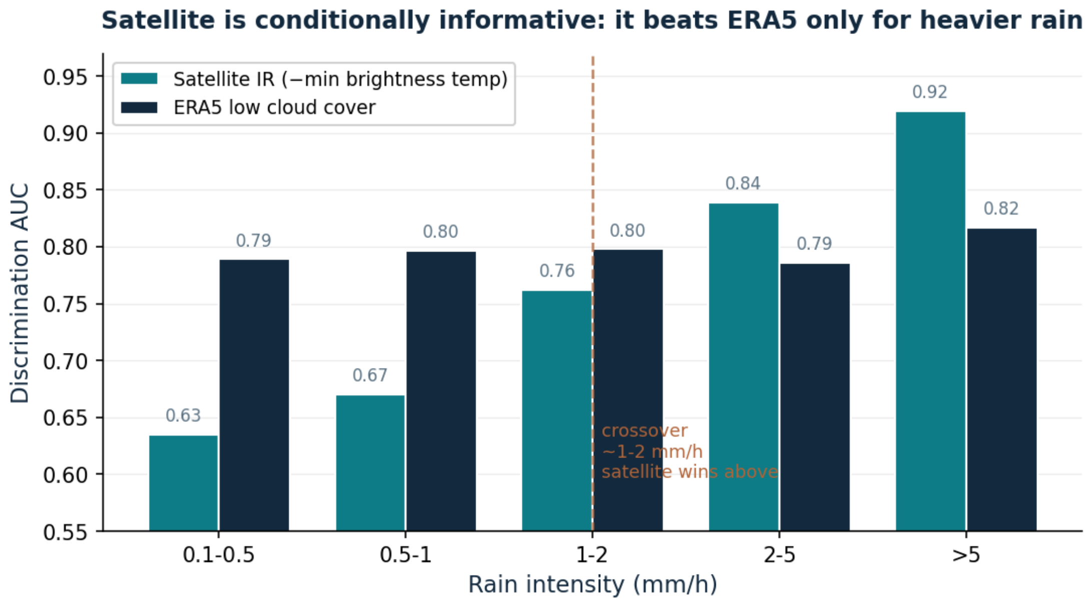

Figure 9. Satellite versus ERA5 discrimination by rain intensity. The satellite carries information the reanalysis lacks, but only for convective rain above ~2mm/h; below the ~1-2 mm/h crossover it is markedly weaker.

The crossover sits at roughly 1-2 mm/h. Above it the satellite clearly beats ERA5 and contributes genuine information; below it the satellite is far weaker. The decisive fact is that 74.4% of test positives are at or below 1.0 mm/h; the exact light-warm-rain regime where infrared retrieval is physically blind[37]. The target's 0.1 mm threshold thus concentrates the label mass in the satellite's weakest regime, so its real contribution (confined to the ~17% of heavier rain events) is invisible in an aggregate PR-AUC and fusion nets out flat. This is a conditional, mechanistic, physically-grounded account, and it ties directly back to the target

[37] Kidd, C. & Levizzani, V. (2011). Status of satellite precipitation retrievals. Hydrology and Earth System Sciences, 15(4), 1109-1116. Infrared methods infer rainfall from cloud-top temperature and therefore systematically under-detect warm-topped (shallow) rainfall while over-detecting cold, non-precipitating hours.

definition disclosure of Section 4f, i.e., it is the threshold choice that buries the satellite's advantage.

d. A tested negative: temporal context (motion)

The single-frame CNN cannot perceive storm motion. To test whether temporal context helps, consecutive hourly patches were stacked as additional channels and the model retrained for one, two and three frames, with all configurations trained and evaluated on the identical set of samples for which the necessary history exists (an infrared-only design, to isolate motion from the day/night confound of Section 7ei). Temporal stacking did not help; test performance declined monotonically as frames were added.

Frames	Input channels	Validation PR-AUC	Test PR-AUC
1	2	0.4114	0.4034
2	4	0.4129	0.3809
3	6	0.3967	0.3638

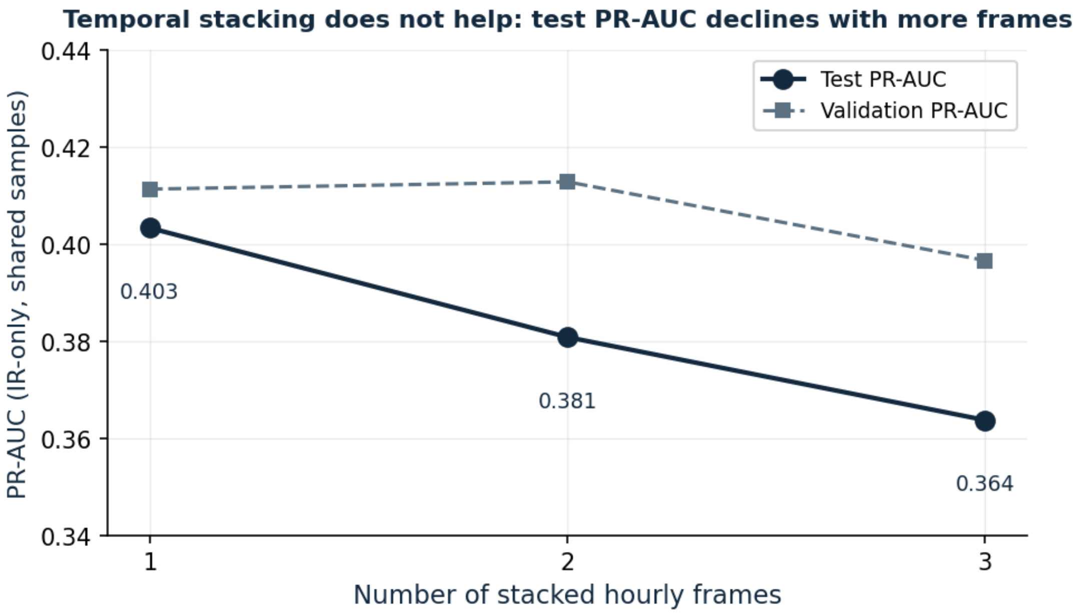

Figure 10. Temporal-stacking test (infrared, matched samples). Adding hourly frames does not improve and slightly degrades test PR-AUC (extra input dimensionality without added signal).

The decline indicates that the extra frames added dimensionality and parameters without usable signal, so the model overfit slightly more. The mechanism is purely physical. Hourly spacing is too coarse for convective motion; at trade-wind speeds a cell crosses the ~128 km patch within an hour, so convective hourly patches are closer to independent snapshots than to a translating field[38]. This is therefore a bounded negative result; hourly temporal context does not help this target, and capturing true motion would require sub-hourly scans

[38] A genuine motion channel would require sub-hourly (e.g. ten-minute) scans, which were outside the downloaded archive.

(a lag sweep independently placed the most informative image one to two hours after the labelled hour[39]), which is identified as the concrete next step in Section 11.

e. Architecture and timing: a recurrent control

Channel stacking discards temporal order: a convolution over stacked frames has no representation of which frame preceded which. To separate that limitation from the sampling interval, a convolutional LSTM was trained[40] on the same three-free infrared sample, carrying a recurrent state across frames instead of concatenating them. A second configuration added, to each frame, its own diurnal (24 h) and semidiurnal (12 h) harmonics as constant input planes, so that any benefit from recurrence had to be information the clock does not already supply.

Configuration	Test PR-AUC
Channel-stacked, 3 frames (Section 8d)	0.3638
Channel-stacked, 1 frame (reference0	0.4034
ConvLSTM, 3 frames	0.4013
ConvLSTM, 3 frames + time harmonics	0.4268

Two results follow. Recurrence recovers most of what channel stacking loses, raising three-frame performance from 0.3638 to 0.4013, so part of the earlier decline was an artefact of the fusion mechanism rather than of the imagery. It does not, however, exceed the single-frame reference of 0.4034, which leaves the central finding intact: three hourly frames add nothing over one. Supplying explicit timing improved the model further, to 0.4268. This was not the anticipated direction for a control, and the interpretation is that time of day is genuinely informative and not recoverable from infrared imagery, which carries no day-night brightness cue. A stratified evaluation confirms the vision signal is not merely a clock: against base rates of 0.035, 0.047, 0.270 and 0.049 for the night, morning, afternoon and evening windows, the model scores 0.4000, 0.330, 0.418 and 0.374 respectively, exceeding the local base rate in every window and by the widest margin in the quiet overnight hours. Carried into late fusion, the stronger vision branch still yields 0.7303 against the tabular model's 0.7557 on the same hours. These figure come from single training runs of a network

[39] The positive lag between surface rainfall and the coldest infrared cloud-top signature reflects the time for a convective updraught to build a mature anvil; it is why the best image-to-label alignment sweep fell one to two hours after the labelled hour rather than at it.

[40] Shi, X., Chen, Z., Wang, H., Yeung, D.-Y., Wong, W.-K. & Woo, W.-C. (2015). Convolutional LSTM network: a machine learning approach for precipitation nowcasting. Advances in Neural Information Processing Systems, 28, 802-810.

trained from scratch, unlike the ImageNet-initialied ResNet, and should be read as indicative of direction rather than a precise estimates.

f. Summary of Model 3

Model 3 is complete. Across five fusion protocols: late (raw and calibrated), feature-level, out-of-fold stacked, and intensity-conditional, no variant beats the tabular baseline (best fusion 0.741 versus Model 1's 0.756). The cause is established mechanically: the satellite is conditionally informative, beating ERA5 only for convective rain above ~2 mm/h, while the target's 0.1 mm threshold places ~74% of positives in the light-rain regime where infrared retrieval fails. Four alternative explanations for the shortfall were tested and eliminated in turn: scale mismatch between modalities, which the standardization already handled; a linear joint head, worth 0.05 when replaced; the dimensional imbalance between embedding and tabular features, worth 0.02 when corrected; and the loss of temporal order in channel stacking, worth 0.04 when replaced by a recurrent architecture. Each produced a genuine improvement and none closed the gap. The temporal work returned a bounded negative, with sub-hourly imagery identified as the path to a genuine motion signal.

9. Final Evaluation and Model Comparison

All three models are evaluated once on the held-out test set, on identical hours (the modality intersection), and reported both including and excluding Hurricane Melissa so that a single extreme event neither flatters nor distorts the picture.

Model	Modality	Test PR-AUC	Recall	Precision
Model 1 – XGBoost	Structured	0.7557	0.841	0.538
Model 2 – CNN (IR + visible)	Unstructured	0.4397	0.847	0.279
Model 3 – best fusion (calibrated late)	Multimodal	0.7406	0.856	0.527

The Melissa effect. Hurricane Melissa (338 test hours, 2.6% of rows, carrying 177 of 1,319 positives) materially lists the headline figure. Reporting Model 1 both ways:

Evaluation window	Model 1 test PR-AUC
Full test set	0.7557
Excluding the Melissa week	0.7089
Melissa week only	0.9515
October 2025 (Melissa's month)	0.8766
All other months	0.664 – 0.783

One week of an extreme, easily classified event is worth roughly +0.047 PR-AUC on the headline. The model's genuine steady-state skill is best represented by the 0.709 figure which excluded Hurricane Melissa, while the full-set 0.756 reflects performance including a major storm the system is precisely designed to catch.

The project's central research question is thereby answered. Using only free reanalysis weather data, the tabular nowcaster reaches 0.756 test PR-AUC (0.709 excluding Hurricane Melissa), roughly five to seven times the random floor. Satellite imagery (costly to acquire and process) adds no discriminative value for this target at this scale, because its genuine contribution is confined to heavier convective rain that the light-rain-dominated target renders invisible in aggregate. This is a useful engineering finding in its own right; a deployable system can run on lightweight weather data alone, without the satellite pipeline's bandwidth and complexity.

10. Deployment Artifact

The deployed system is an inference service that turns the trained model into a live deploy-or-stow decision. On each cycle it takes the current weather conditions and the recent hours of history, applies the exact feature transformations used in model training (same feature engineering module for both training and inference), runs the model, and returns a wind-driven-rain probability together with a binary decision (deploy/stow) at a chosen threshold.

A deliberate design choice governs which model is deployed. The evaluation established that satellite imagery adds no discriminative value beyond the reanalysis weather features (Sections 8-9). The deployed service therefore runs the tabular model alone. This is not a simplification for convenience purposes, rather it is a direct consequence of the finding: *deploying the satellite pipeline would add bandwidth, compute, and complexity for no measured benefit*. Thus, the result is a lightweight service that runs on weather data alone, without a satellite feed or a GPU.

The demonstration interface exposes the system as an interactive console. Rather than requiring users to supply the full engineered feature vector, it presents meaningful, human understandable levers (location, time of day, cloud cover, humidity, recent rainfall, wind gusts, and the pressure trend) and derives the remaining features internally. The decision threshold is exposed as a control, making explicit that the deploy/stow cut-off is an operating point decision governed by the relative cost of a missed wetting versus an unnecessary deployment, rather than a fixed property of the model. A second mode replays genuine hours from the held-out test set, showing the model's decision alongside the recorded outcome, so that accuracy on unseen real data can be inspected directly.

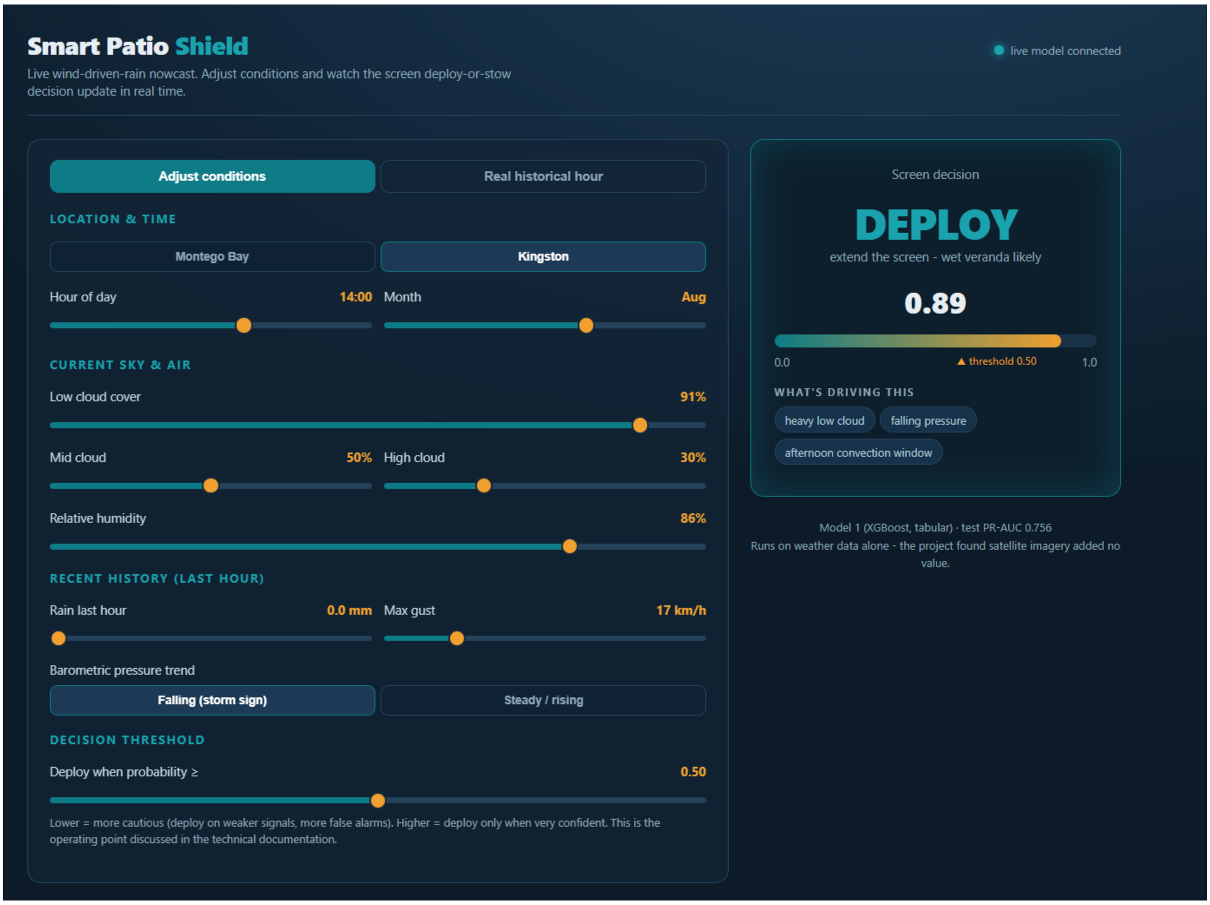

Figure 11. The Smart Patio Shield inference console. The left panel exposes the meaningful weather levers; the right panel shows the live deploy-or-stow decision, the predicted probability against the chosen threshold, and the factors driving the call.

The demonstration derives the unexposed features from the exposed ones, so it is a faithful interface to the model rather than a live weather feed. A production deployment would instead pull real-time observations from the Open-Meteo API and compute the lag features from the genuine preceding hours. The decision threshold is set at 0.5 by default; its operational value would be tuned to the relative cost, for a given user, of a missed wetting against an unnecessary deployment.

11. Limitations and Future Work

a. Documented Limitations

- **Target is effectively a rain detector.** At the 15 km/h gust threshold the wind clause does almost no work; the target agrees with a plain rain detector on 99.09% of hours (Section 4f). The thresholds are provisional and uncalibrated against observed ground truth.
- **Vision margin is modest against the right baseline.** Model 2 exceeds a climatology lookup table by only +0.12 PR-AUC, not the +0.34 its margin over random suggests (Section 7d).
- **Visible-band gain is partly a day/night proxy.** The +0.023 channel-ablation gain cannot be cleanly attributed to cloud reflectance versus a time-of-day signal (Section 7e).
- **Satellite is conditionally, not universally, informative.** It beats ERA5 only above ~2 mm/h; the target buries that regime, so fusion shows no aggregate gain (Section 8c).
- **Fusion protocol caveats.** The meta-learner was fit on fewer hours than the base models saw; out-of-fold stacking was clean on tabular branch but in-sample on the vision branch (no k-fold CNN retrain); validation and test cover different seasons.
- **Temporal context untested below one hour.** Hourly stacking did not help (Section 8d); sub-hourly motion was not acquired.
- **Instrument shift and non-determinism.** Train is entirely GOES-16 and test entirely GOES-19 (a *1.37 K mean C13 shift, ~0.05 std, judged negligible); GPU training is not bit-reproducible (key results verified stable to ±0.002).

b. Future work, in priority order

- **Sub-hourly imagery for a true motion channel.** Stacking three ten-minute scans within the labelled hour is the concrete, evidence-backed next step. It is better temporally aligned than the single scan and would supply the motion the hourly test could not. A recurrent architecture is now the appropriate vehicle for this: the convolutional LSTM tested here recovers the ordering information that channel stacking discards, so it is the natural model to apply once finer-grained scans are available.
- **Recalibrate the target.** Calibrate the gust and precipitation thresholds against an observational log so the wind dimension genuinely binds, then re-evaluate each model.
- **Daytime-only vision evaluation** to separate the visible band's cloud signal from its time-of-day proxy.
- **Intensity-targeted deployment.** Since the satellite wins above ~2 mm/h, a two-stage design that consults imagery only for suspected heavy-rain hours may capture its value without the aggregate dilution.
- **A true T+1 forecasting formulation,** under which anticipatory features and imagery carry more weight than they do in the present nowcasting setup.

12. Conclusion

This project delivered a reproducible, end-to-end multimodal pipeline for nowcasting wind-driven rain in Jamaica, and an unusually rigorous account of what the data can and cannot support. The tabular model is a strong nowcaster (test PR-AUC 0.756, or 0.709 excluding Hurricane Melissa) built entirely on free reanalysis data. The satellite vision model extracts real but modest information (+0.12 over a climatology baseline), limited by infrared blindness to the war, light rain that dominates the target. Multimodal fusion did not beat the tabular baseline under any of the five protocols, and the reason was established mechanistically rather than asserted: the satellite is conditionally informative, contributing only for heavier convective rain that the target's threshold renders a small minority of events. A temporal motion test returned a bounded negative and identified sub-hourly imagery as the concrete path forward.

The scientific value of the project lies as much in its negative results as its positive one. Each was reached by a controlled experiment and explained by physics: the persistence decomposition, the climatology baseline, the intensity-stratified satellite comparison, the five-variant fusion analysis, and the motion test together form a coherent, honest picture of a hard prediction problem. The headline engineering conclusion is actionable (the deployable product can run on lightweight weather data alone) and the research conclusion is defensible: at this scale and with this target, satellite imagery does not improve wind-driven-rain nowcasting, for a reason that is precisely understood.

13. References

Sources are cited in footnotes at the point of use and collected here alphabetically.

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14. Appendices (.py files → notebooks)